Name: ____________

Date: ____________

Function Operations, Composition & Inverses

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can determine whether a function is even, odd, or neither.**

RELATED SKILLS

- D — Test a formula or graph for even or odd symmetry.

☐ **2. I can add, subtract, multiply, and divide functions and state the resulting domain.**

RELATED SKILLS

- I — Add, subtract, multiply, and divide function outputs.
- I — Determine the domain of a sum, difference, product, quotient, or composition.
- A — Simplify an algebraic expression using operation properties.

☐ **3. I can evaluate a composition from formulas, graphs, or tables.**

RELATED SKILLS

- D — Follow intermediate values to compose functions from tables.
- D — Substitute one function formula into another.
- A — Identify input and output values in a formula, graph, table, or context.

☐ **4. I can write a formula for the composition of two functions and determine its domain.**

RELATED SKILLS

- D — Determine the domain of a sum, difference, product, quotient, or composition.
- D — Substitute one function formula into another.

☐ **5. I can decompose a function into a composition of simpler functions.**

RELATED SKILLS

- I — Identify simpler inner and outer functions in a composition.
- A — Substitute one function formula into another.

☐ **6. I can determine whether a function has an inverse on a given domain.**

RELATED SKILLS

- D — Apply the horizontal line test for invertibility.
- R — Test whether a function is one-to-one on a domain.

☐ **7. I can find and verify an inverse function algebraically.**

RELATED SKILLS

- D Exchange variables and solve to obtain an inverse formula.
- D Verify inverse functions using composition.
- A Substitute one function formula into another.

☐ **8. I can interpret the relationship between a function and its inverse from graphs or tables.**

RELATED SKILLS

- D Exchange domain and range between a function and its inverse.
- D Exchange input and output values when interpreting an inverse.
- D Reverse input and output units when interpreting an inverse.